Paper Review | arXiv:2611.01112

Auriga: Zero-Shot Embodied Manipulation with a Task-Agnostic Agent Harness

Auriga Technical Report: A Task-Agnostic Agent Harness for Zero-Shot Embodied Manipulation

One-Sentence Summary

No policy training, no trial-and-error on target tasks: wrapping a general VLM in a "**preview, execute, stop on deviation**" harness closes **68.3%** of the planning–execution gap under single-episode zero-shot conditions and tops RoboDojo (**57.7% SR**).

First submitted 2026-10-31 • Base model: GPT-6 Astra (high)

Fig. 1: one episode—locate→preview→miss caught→re-grasp→done

TL;DR

Auriga in Four Numbers

57.7%

#1 on RoboDojo

64.0 Score, 6,300 episodes; reference harness with the same model: only 22.5% SR / 29.0 Score

68.3%

Plan–Exec Gap Closed

Reference interface 22.5% → arm-free upper bound 76.2%; Auriga reaches 59.2% (210-episode diagnostic set)

45.6%

#1 SR, Randomized

Standard 52.3% → randomized 45.6% (−6.7); learned policies drop by up to 33.9 points

73.2%

RoboCasa365 Overall

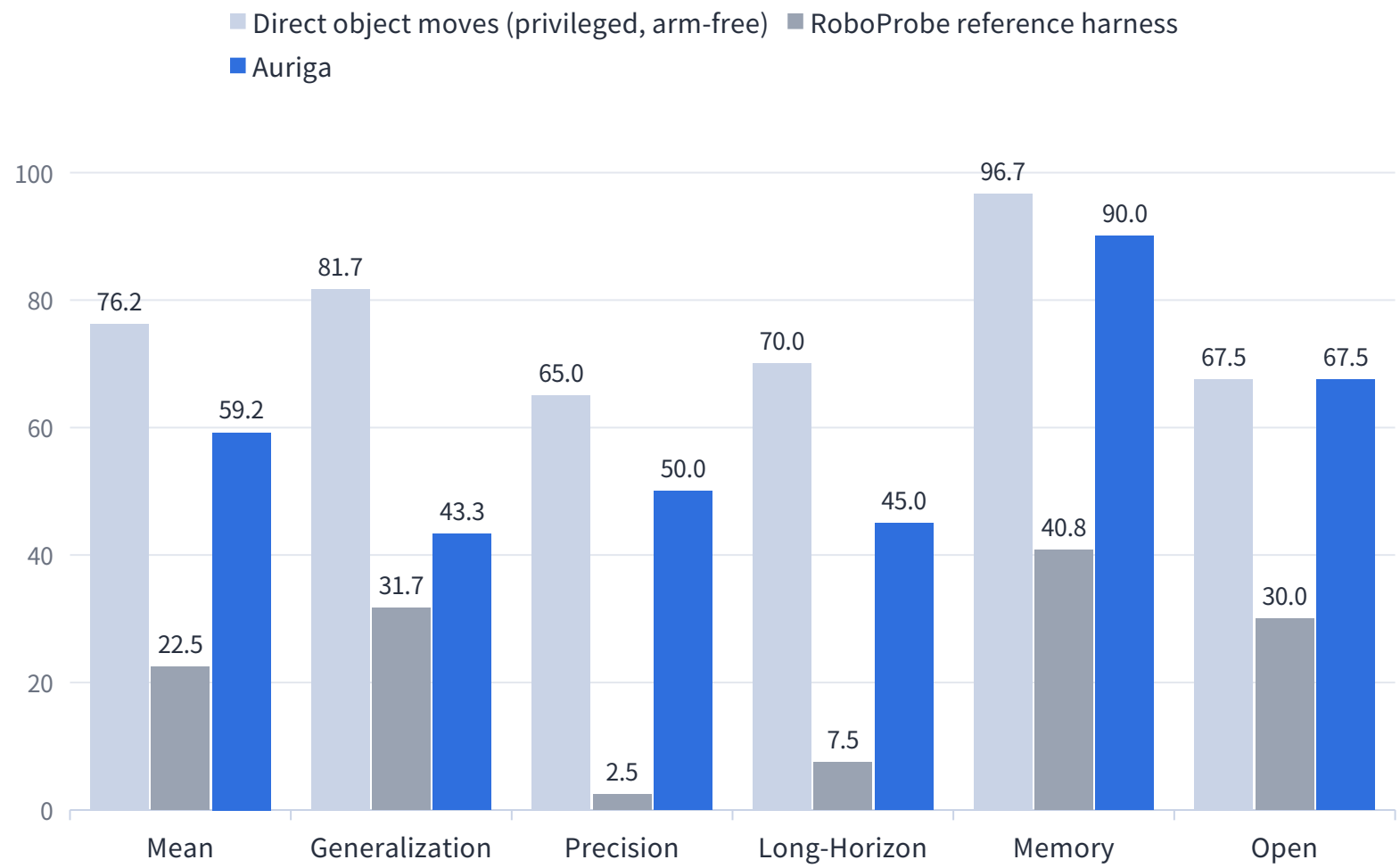
No VLA, no cross-episode memory; leads VLA agents by 19.4–23.1 points on composite tasks

Method Keywords

- **Task-agnostic:** no task-specific code/prompts/skills, no privileged sim state; frozen for evaluation after 13 cross-task iterations
- **Model makes all decisions:** one persistent session + explicit task belief B_t (plan/focus/premise ledger) + offline evidence log E_t
- **Single admission gateway $\mathcal{A}$:** preview the full trajectory in a paused world → replay verbatim → verify per-stage tolerances; stop on deviation and return the measured error

01 Problem

The bottleneck is execution, not planning: right plan, failing arm



Success rate SR (%); RoboProbe Memory column based on 29 episodes

Gap Closed

68.3%

(59.2 − 22.5) / (76.2 − 22.5)

70.6% by Score

- The model usually picks the right object and destination
- Failures come from execution: grasps off by a few mm, drops in transit, hitting container rims
- Common control interfaces fail to catch these errors in time

01 Problem

Prior paradigms leave the gap; Auriga closes it under stricter zero-shot rules

Where are execution judgments made?

Stepwise / chunked actions
Reference harness, VLA

Issues one step at a time with no view of the full path; collisions found only after the arm moves

Code-as-Policy
Programs call the control API directly

Scripts can't see the scene; a program may exit 0 even when an action fails

Auriga
Full sequence + preview + verification

Before execution, inspect the full trajectory, duration and clearances; verify measured results after each stage

Single-Episode Zero-Shot Protocol

1. **No target-task exploration before evaluation**

   No prompts, programs, skills or policies developed on target tasks (incl. other layouts); model weights fixed

2. **No demonstrations at evaluation**

   Only the task description and permitted observations

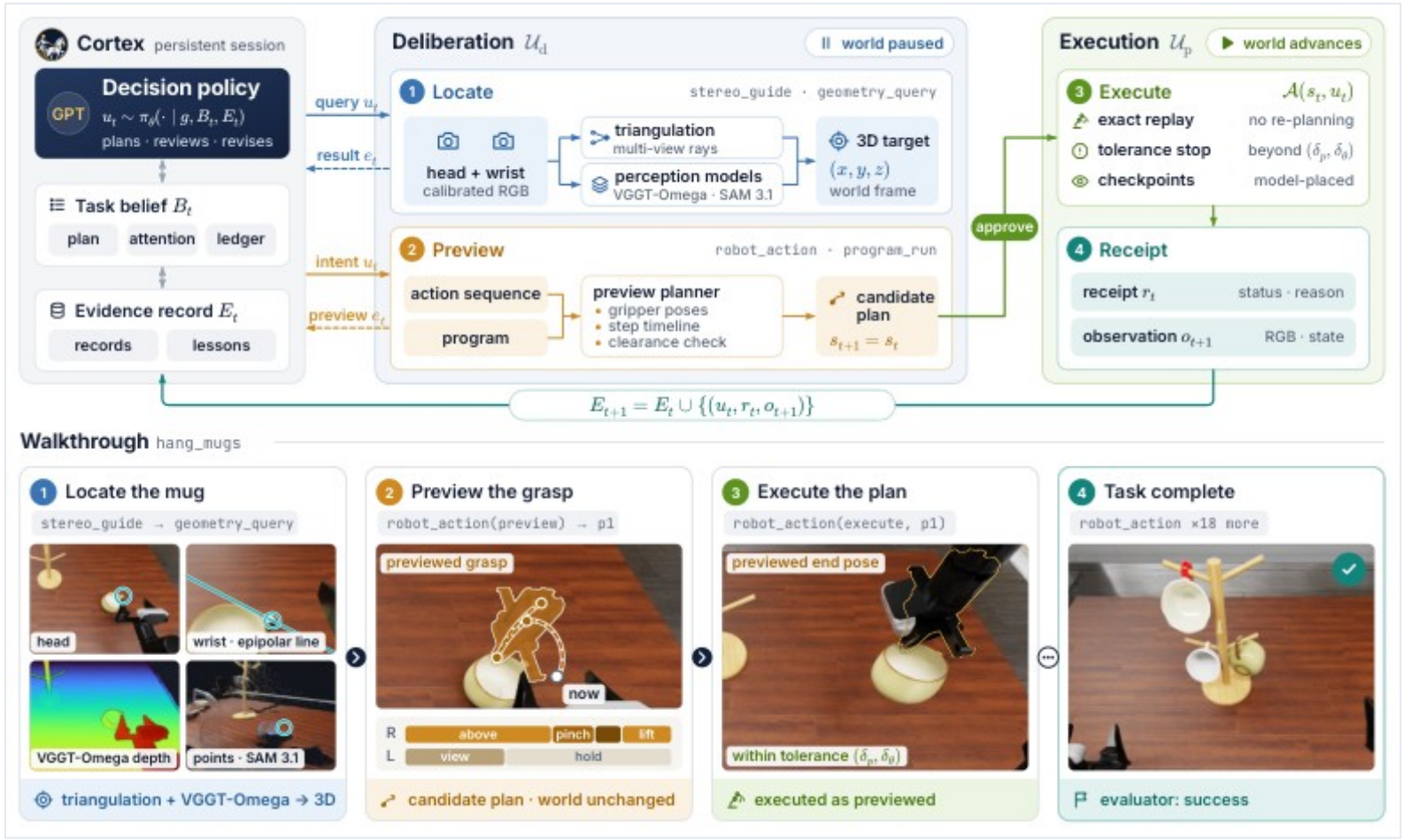
3. **Each case executed only once**

   No resets or reruns; recover in place from the current physical state

Contrast: HexaShell / EmbodiedEvo / SEAR repeatedly reset and tune in target scenes (RSI L0–L2).

Auriga: failures drive upgrades to general localization, contact-checking and state-tracking tools (L3–L5), frozen at evaluation.

Three layers: model decides, tools measure, one gateway admits motion



Paper Fig. 4: Auriga overview (top) and closed-loop hang_mugs execution (bottom)

Cortex • Persistent model session

Makes all task decisions (decompose, measure, set goals, recover); no task-specific prompts

Capability layer • General tools

Measure / Act / Compute / Recall: provide capabilities, never encode solutions

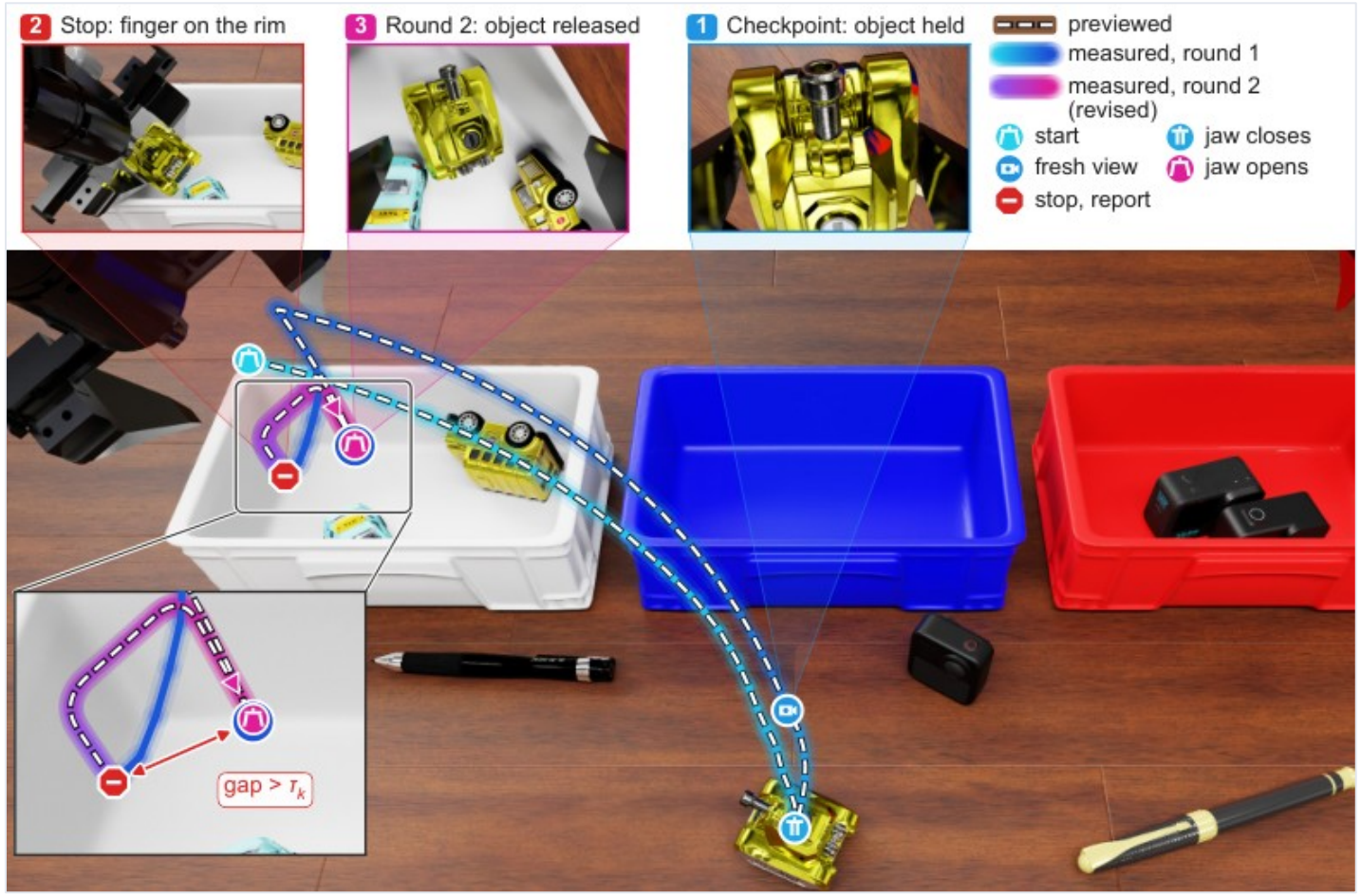
Admission gateway $\mathcal{A}$

The only channel that moves the robot; direct calls and program requests share one path

Deliberative calls $\mathcal{U}$d: s_{t+1}=s_t; world paused, no sim time consumed

Physical calls $\mathcal{U}$p: (s,r,o)=$\mathcal{A}$(s,u); must pass the gateway

Preview–execute–verify: replay exactly; halt & report drift



Paper Fig. 6: two rounds of preview and verification (basket-rim contact and round 2 illustrative)

① Preview — World paused

Full multi-stage bimanual sequence → projected trajectory, steps per stage, clearance warnings; not executed if unplannable

② Verbatim Replay — No replanning

Executes exactly the previewed trajectory; free-space deviation ≈ contact signal

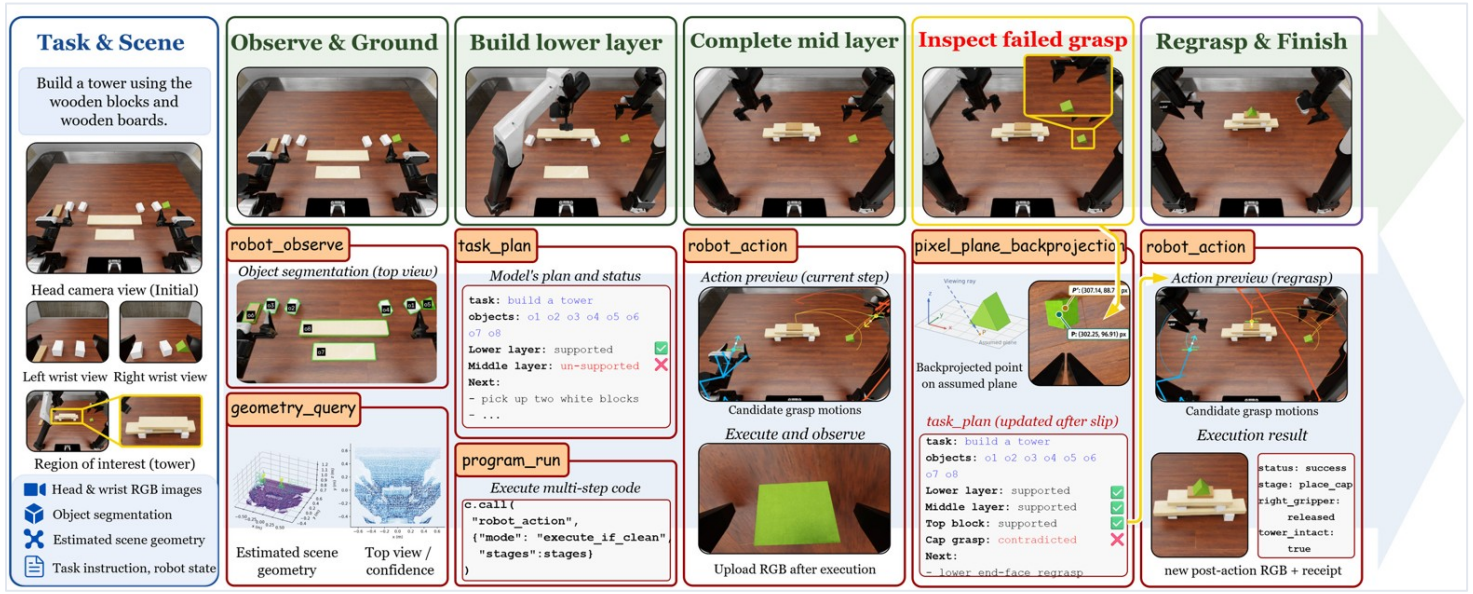
③ Verify — $\tau k=(\delta p, \delta\theta)$

Default 10 mm / 0.15 rad; if exceeded, halt and hold; receipt gives failed stage, deviation, unexecuted stages and new views

④ Checkpoints — Model-placed

Unconditional or triggered by gripper aperture; outperforms “pause after every gripper action”

Explicit task belief + three-level loop: recovery keeps completed progress



Paper Fig. 2: planning, execution and recovery in a tower-building episode (scene never reset)

swap_T example: action "done" but gripper aperture 0 → blue block "held" marked contradicted, pink placement still supported → switch to left arm to fix only the blue block, without redoing the whole swap

Task belief B_t (model-maintained)

plan — Goal, method, completion criteria

attention — Key current uncertainties and evidence that resolves them

ledger — Key premises: pending / supported / contradicted / unknown

Three-Level Loop

Stage loop Gateway $\mathcal{A}$

Tol./grasp-check fail → halt before next stage

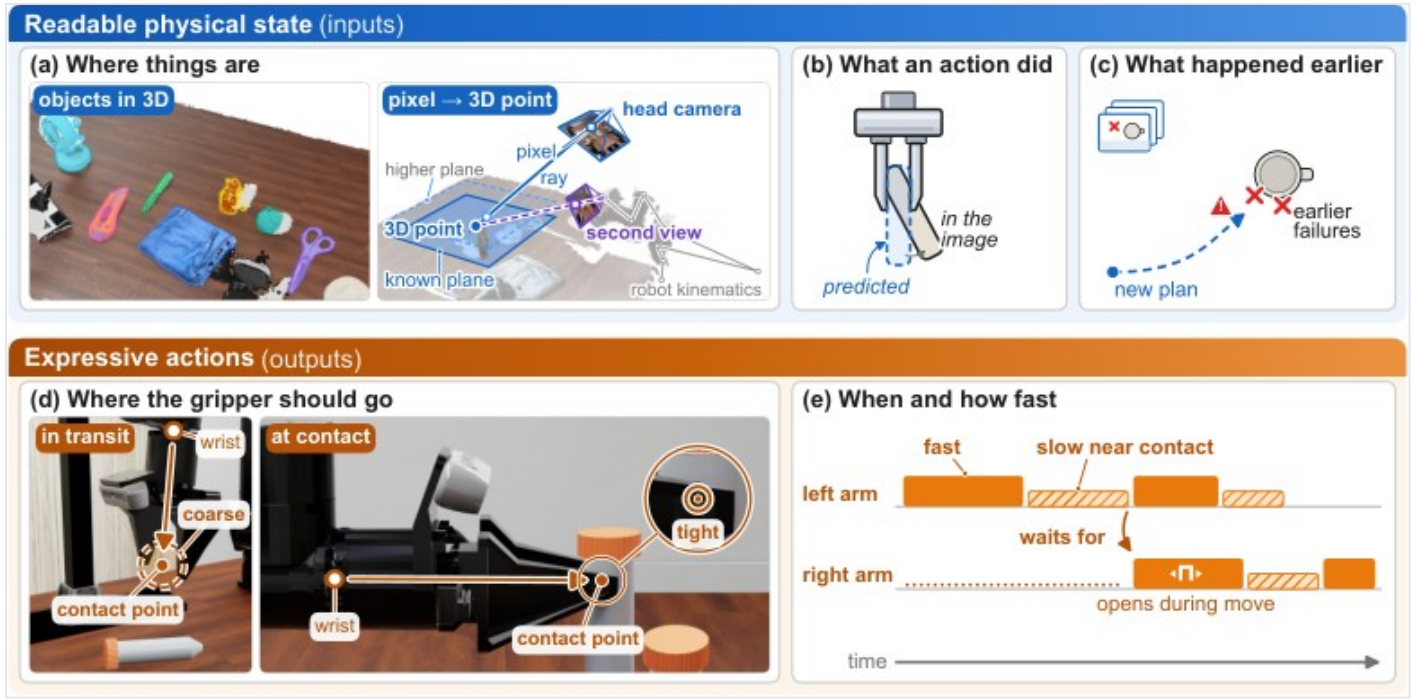
Action loop Model

Receipt vs. new obs → cause hypothesis → act

Belief loop Model

Repeated refutation → new contact/route/goal

Turn physical state into readable facts; express actions in task language



Paper Fig. 7: perception and action interfaces (a–c input side, d–e output side)

Input: Readable Physical State

- Pixel ray $d = R\,K^{-1}\,(u, v, 1)^{T}$; 3D localization = solving for distance along the ray
- Coarse: VGGT-Omega depth (scaled by table height) + SAM 3.1 segmentation → per-object pose/size/reliability
- Fine: intersection with a known-height plane (reports $\partial p / \partial h$) or two-view wrist-camera triangulation
- Measured gripper aperture and in-hand slip; warnings when revisiting past failure regions

Output: Expressive Actions

- Targets specified at fingertip/finger-pad contacts, not the wrist flange
- Per-stage tolerances: loose for transport, tight for contact
- Shared bimanual timeline: declarable waits, slow-down near contact, gripper actuation while moving

13 cross-task iterations: 21.5 → 64.0 by upgrading general tools

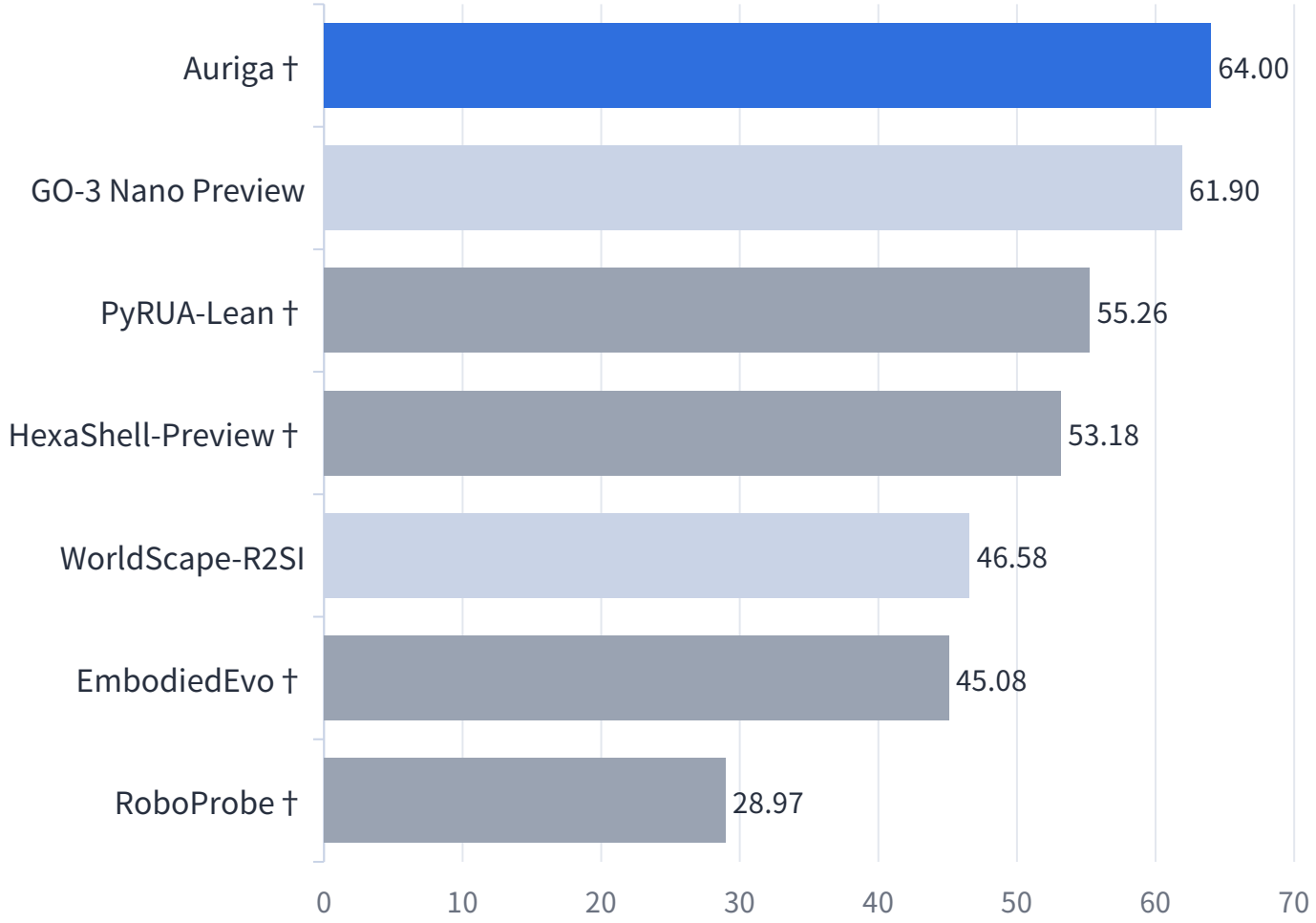
Milestones (Overall Score)

#2	Motion preview	26.2
#3	Unified bimanual actions	30.0
#4	Refactor + verbatim task text	45.5
#7	ROI zoom	49.9
#9	Stage tol. + obstacle check	53.4
#11	Explicit motion control	58.8
#12	VGGT-Omega + SAM 3.1	64.0

#4 → final: Long-Horizon +29.5, Memory +30.8, Precision only +5.6

Paper Fig. 3: harness capability iterations #0–#12 (RoboDojo overall score)

#1 on RoboDojo: 64.0, 2.2× the reference harness with the same model



Mean Score (0–100); † uses gpt-6-astra; light = WAM/VLA learned policies (incl. agent combos)

64.00

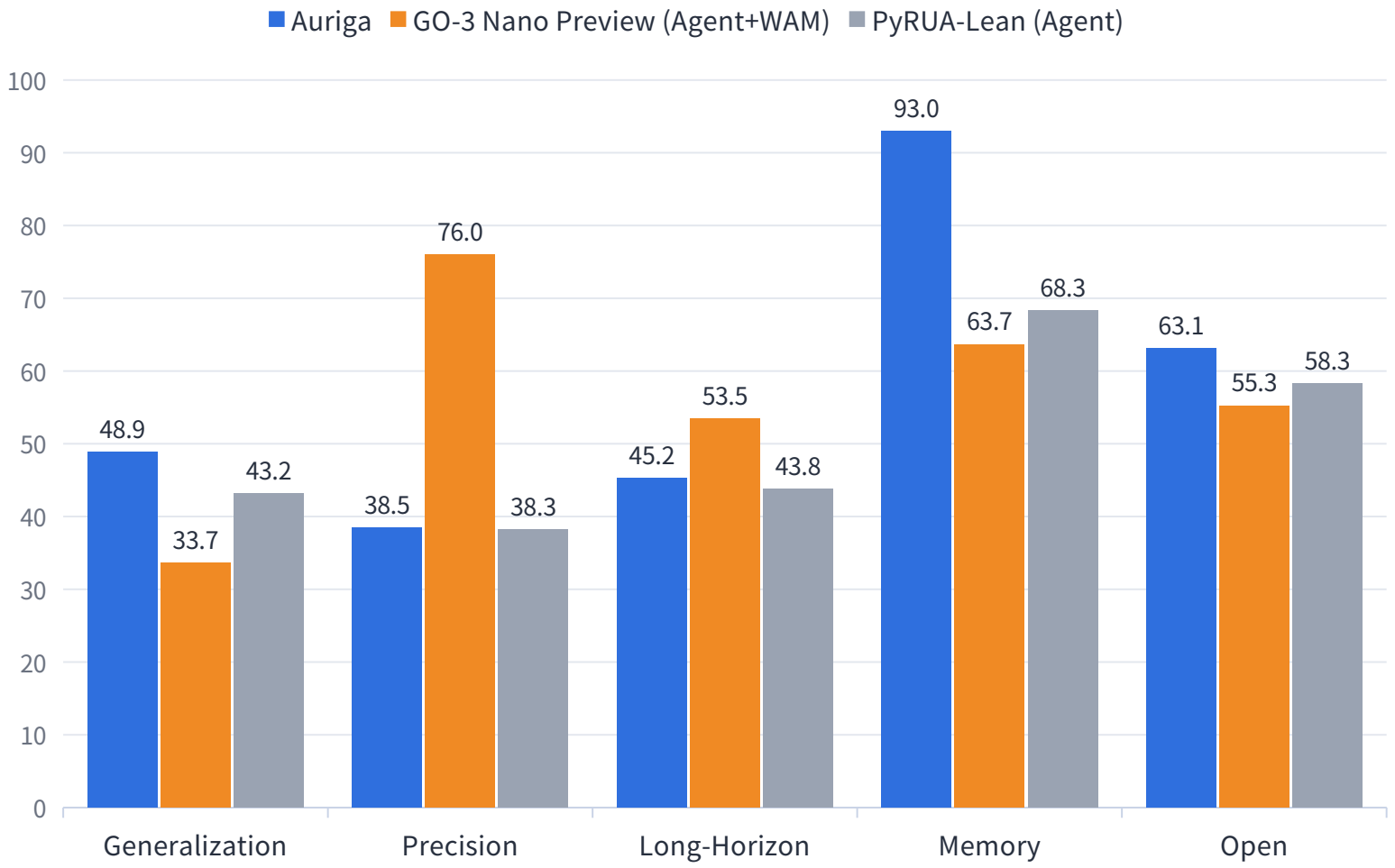
Mean Score

57.74%

Mean SR

- **vs RoboProbe (same model):** 28.97 → 64.00 Score, 22.48% → 57.74%
- **vs GO-3 Nano (Agent+WAM):** 61.90 / 56.42%, #2 overall
- **Agent group:** #1 in all 12 columns; 8.7 points ahead of PyRUA-Lean
- **Per-task Score:** 24 wins / 17 losses vs PyRUA-Lean; 41 wins / 1 loss vs RoboProbe

Leads in Memory, Generalization, Open; clearly trails GO-3 in Precision



Per-dimension SR (%), data from Tab. 4

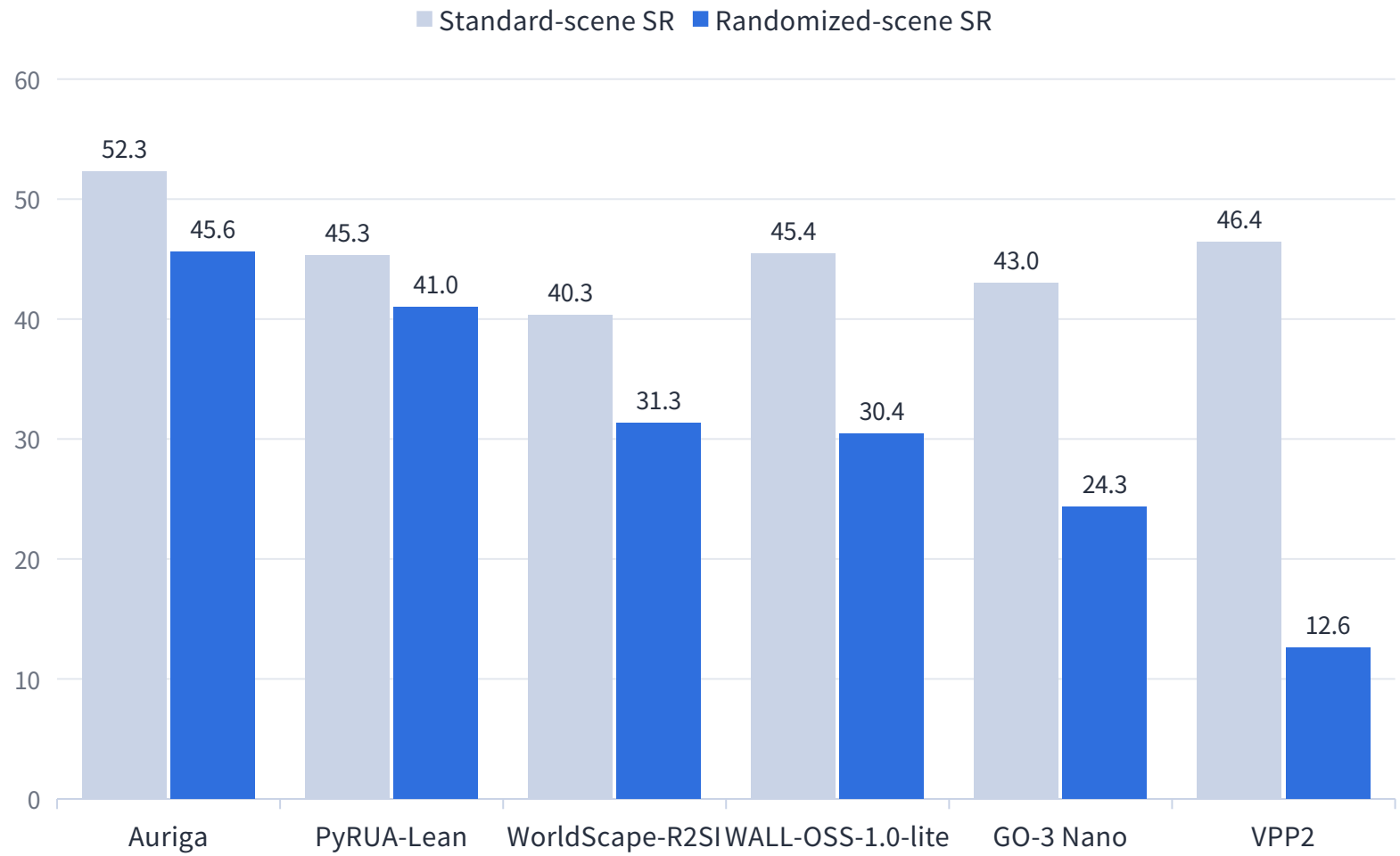
Strengths

- Memory 93.0% (+12.0 points)
- Generalization 48.9% （+5.7）
- Open 63.1% （+4.8）

Weaknesses

- Precision 38.5% vs GO-3 76.0%
- Long-Horizon 45.2% vs 53.5%
- insert_key, store_tools_in_toolbox, store_laptop_and_headphones all at 0

Still #1 after scene randomization; learned policies lose up to 33.9 pts



SR (%), sorted by composite generalization metric G (selected methods)

Randomized − Standard (SR, pp)

Auriga	−6.7
PyRUA-Lean	−4.3
WALL-OSS-1.0-lite	−15.0
GO-3 Nano Preview	−18.7
PhysicalRSI	−20.1
VPP2	−33.9

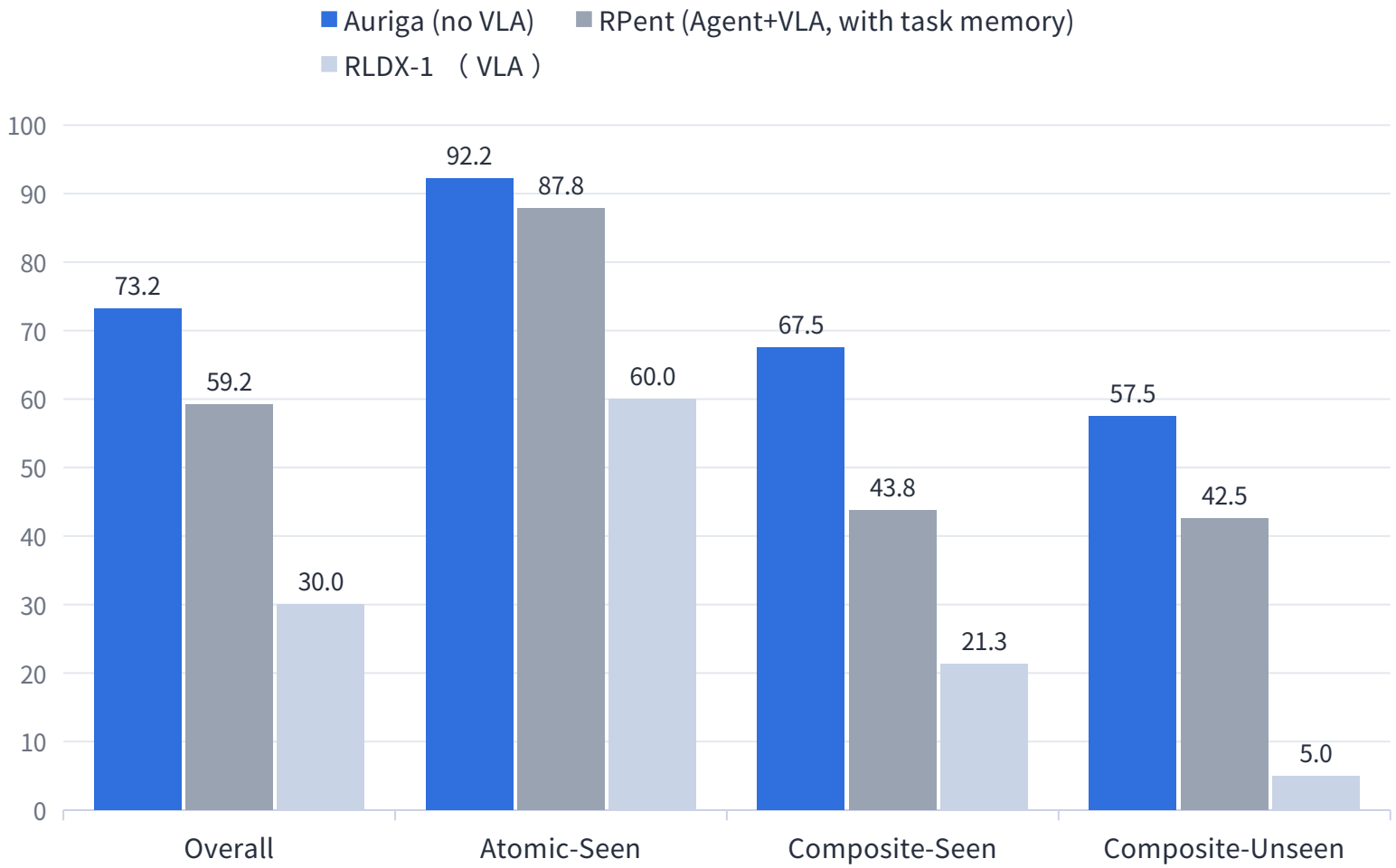
Composite Generalization G

46.27

#1; PyRUA-Lean 43.87 is #2

$G = \frac{1}{2}(S_{std}+S_{rand}) - |S_{rand}-S_{std}|$

RoboCasa365 transfer: 73.2% without VLA; biggest gain on composite tasks



62.5%

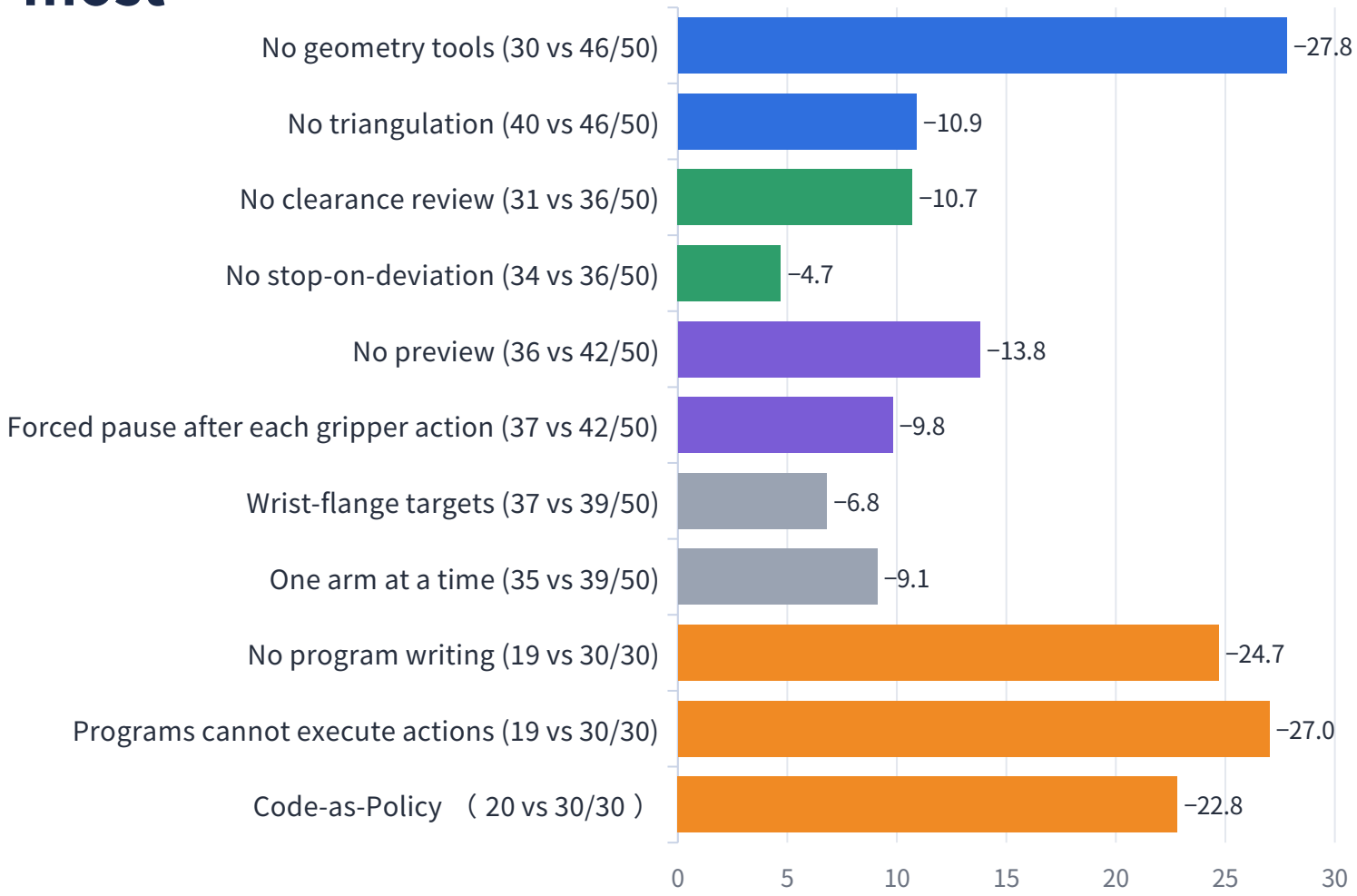
Composite Overall SR

+19.4 / +23.1

- Baselines hand contact-rich stages to RLDX-1, which alone scores only 5.0% on Composite-Unseen
- Auriga completes every stage with stage-verified action programs
- 67 failures: 23 exhausted requests, 44 where the model stopped acting
- Note: this environment provides rendered depth-coordinate maps and live predicate values

SR (%), 250 episodes, all in unseen target kitchens; PyRUA-Lean reports only Overall 56.4 / Composite overall 39.4

Ablations: 3D measurement and “reviewable programs” contribute most



Mean Score drop vs. same-group Full; parentheses: successes (ablation vs Full)

Silent Failures of Code-as-Policy

Of 120 program-issued actions, 2 were halted for exceeding tolerance and 1 paused at a checkpoint;

without receipts, all these programs would return **exit 0**

and one even issued 8 more actions.

Full: 30/30, Score 100, 46.2 model calls (vs ~66 −72)

Cost: more calls and tokens; failed episodes consume ~70% of input

Per-episode mean	Auriga	PyRUA-Lean
Mean Score	**64.00**	55.26
Model calls	**47.5**	25.7
Input tokens (M)	**5.63 (85.4% cached)**	0.75
Output tokens (k)	**18.3**	9.0
Duration (min)	**18.8**	12.7

Tab. 11; both at reasoning effort high

Successful episodes (3,405)

30.3 calls

3.10M input • 11.5 min

Failed episodes (2,894)

67.6 calls • 27.5 min

70.2% of all input tokens

swap_T matched scenes: Auriga vs RoboProbe

14/15

Auriga success

RoboProbe 3/15 (+73.3 points)

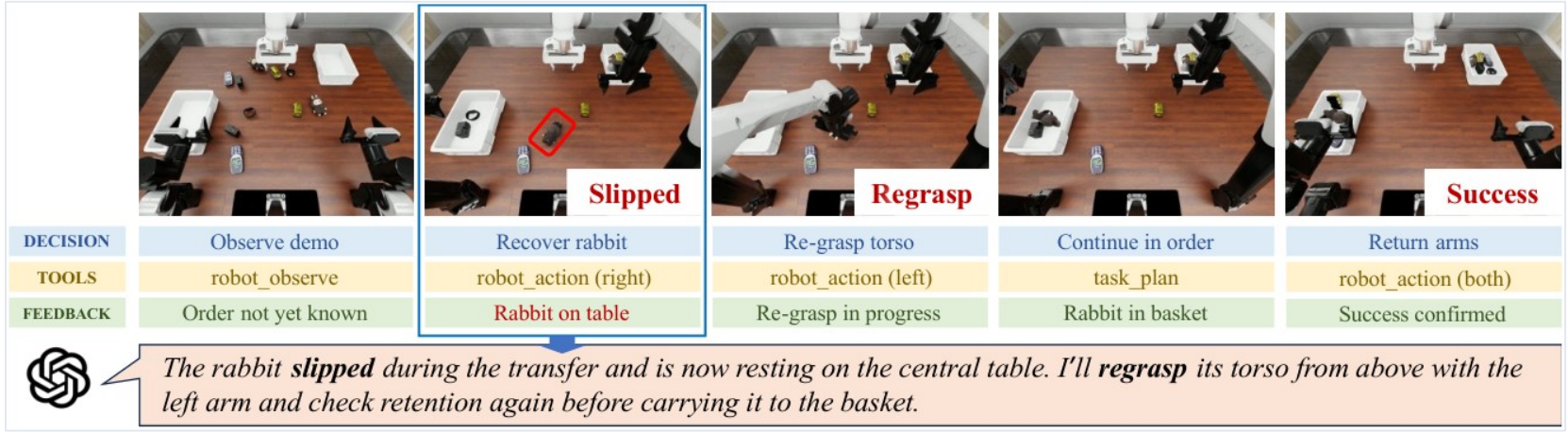
−90.6%

Execution Batches

3.1 vs 33.4 per episode; 107.3 vs 10.9 steps per batch

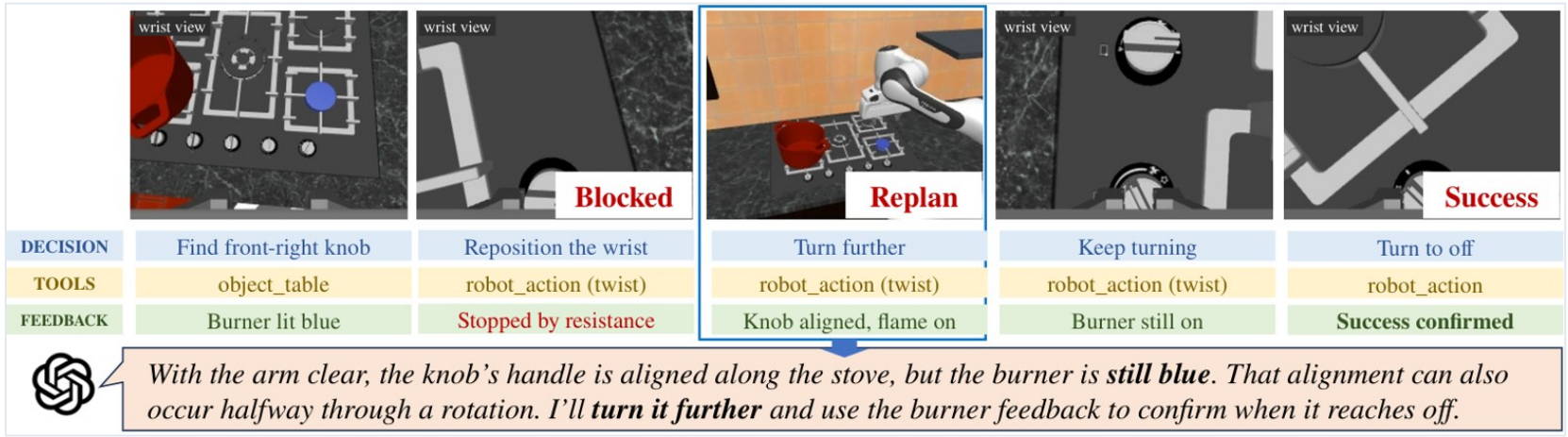
- Fewer model calls: 25.4 vs 36.6 per episode
- Similar total steps: 336.2 vs 365.1
- Approach–grasp–transport–release previewed and verified as one multi-stage operation

Qualitative: diagnose via receipts; switch grasp, arm or strategy



imitate_sorting_sequence

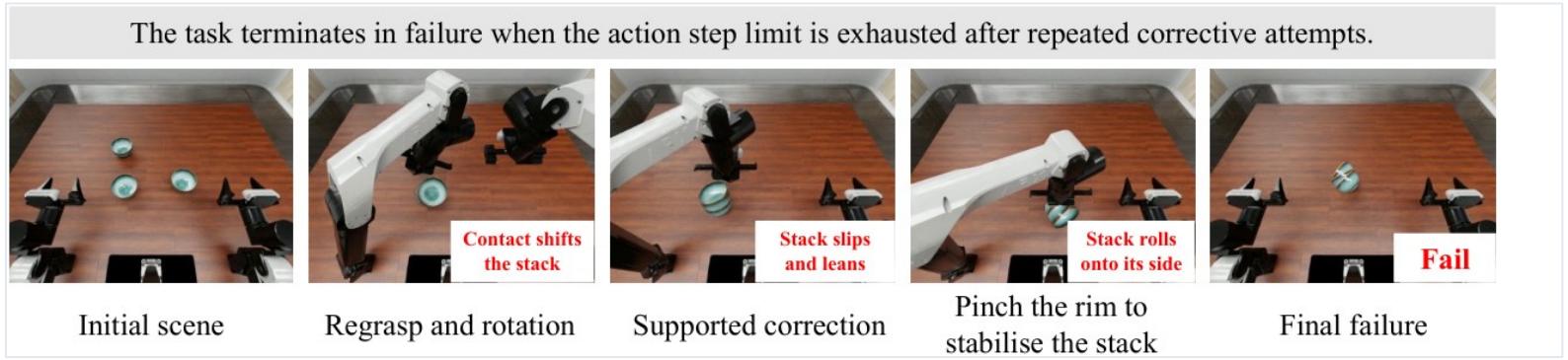
Bunny slips during transfer → slot 3 stays pending, no skipping ahead → left arm grasps the torso → hold confirmed → finished in order



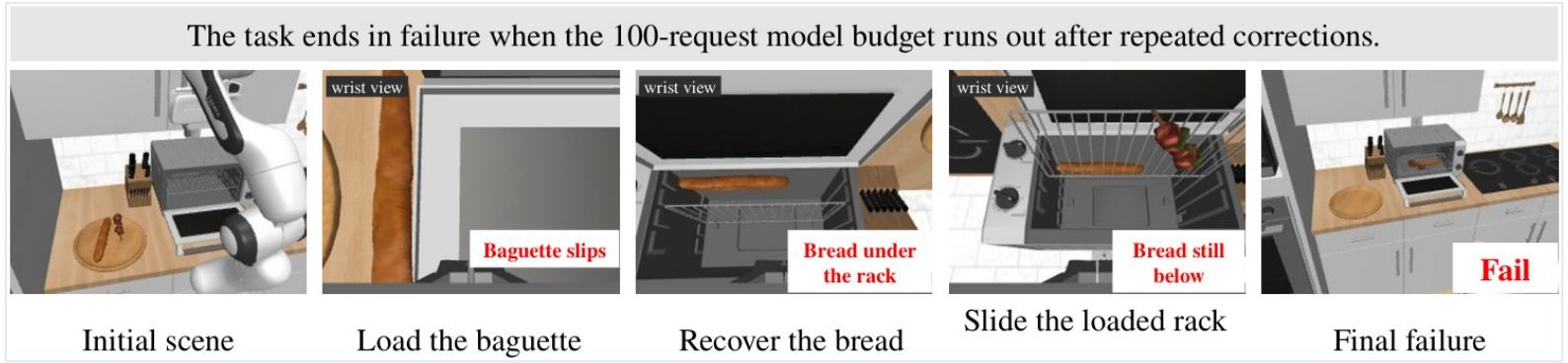
TurnOffStove （RoboCasa）

Rotation blocked → release and adjust wrist pose; knob looks "off" but flame still blue → keep rotating, trusting the flame state

Failure mode: re-fixing coupled multi-bodies drains budget



(a) RoboDojo stack_bowls: 800-step budget exhausted, Score 15



(b) RoboCasa365 HeatKebabSandwich: 100 requests exhausted

stack_bowls （Tab. 22）

Step 242: two bowls stacked

Step 440: righting the top bowl lifts the whole stack, which slips and topples

Steps 691→800: topples again after arm retracts; budget exhausted

HeatKebabSandwich （Tab. 23）

Request 41: baguette falls under the grill

Requests 63→86: repeated slips on the thin front rail

Request 100: requests exhausted, baguette still under the rack

Key lesson: deviations are reliably detected, but with coupled multi-body structures or secondary obstacles, a locally “completed” action does not mean overall progress.

Takeaway: solid out-of-model execution loops may beat a new VLA

Contributions

- **Diagnosis:** quantifies the "planning–execution gap" with arm-free privileged actions, locating the problem in the execution loop
- **Design principle:** the harness only measures and halts to report; all decisions stay with the model
- **Unified gateway:** programs and direct calls share one preview→replay→verify channel, preventing silent failures
- **Results:** #1 on RoboDojo under strict zero-shot (57.7%), robust to randomization, cross-benchmark transfer (73.2%)

Limitations & Open Questions

- **Simulation only:** paused-world deliberation + 18.8 min per episode do not hold under real-robot real-time constraints
- **Weak fine contact:** Precision 38.5% vs GO-3 76.0%; no force control
- **Benchmark-driven development:** all 13 iterations measured on all RoboDojo tasks; the RoboCasa environment gives more privileged information
- **Cost & dependency:** ~7.5× PyRUA-Lean's input tokens; only GPT-6 Astra tested
- **Ablation scale:** 10 tasks, 30–50 cases per group; limited confidence

In one sentence: with a solid execution loop of measurement, preview, verification, evidence and explicit task state, a general VLM can greatly raise simulated manipulation success without specially trained policies; next come force-sensitive fine contact, real-time constraints and recovery in coupled multi-body scenes.